L01: Data Driven Problem Solving

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January 5, 2026

About the course: Scope

- A **fundamental** course on machine learning.
- **Not an** advanced course on ML/AI
- Emphasis on:
  - Part I: Basics Concepts, Principles, Basic Maths, Basic Algorithms
  - Part II: Intro. to Neural networks and deep learning.
- Note: Class could be highly heterogeneous:
  - UG, PG, Working Professionals, New Students, People who have taken an ML course else where, students who use these ideas in their own research/work etc.
- Regular work is expected; What does it may imply?
  - Lecture/class is for learning, interacting
  - Regular grading. Get graded while learning.
  - In-class activities;
  - Communicate and discuss; Learning is the focus
- Course accounts buffers to take care of unfortunate network failures. Minimize special requests

About the course: Getting Ready

- Exposure/Background:
  - Maths in the form of UG Courses and Schools.
  - Programming/Algorithms: Comfortable with adaptation/hand-on.
- Maths:
  - Typical Engg Maths. Not super advanced.
  - Topics: Linear Algebra, Probability, Differential Calculus
- Programming:
  - Most tools/libraries are in python. No plan to teach python or programming. numpy etc. for coding mathematical operations.
  - No super-heavy compute expt. planned. Some TA support will be provided.
- Infrastructure:
  - Some compute, say a laptop.
  - Internet connection (say 4G/5G) expected.

About the course: Course Coverage

1. **Part I: Foundations of Machine Learning**
   - Mathematical Foundations; Role of Linear Algebra and Probability; Supervised Leaning Formulation and Challenges, Sample Algorithms; ML problem formulation; Role of Data Linear Methods in Machine Learning; Regression, PCA, Regression, Perceptrons, Gradient Descent, Multiclass. se;ected additional topics.
   - Total of 11-12 Lectures
2. **Part II: Introduction to Modern Machine Learning (Neural Networks)**
   - Artificial Neural Networks, Perceptrons, Multi Layer Perceptrons, Gradient Descent and Back Propagation; Intro to Deep Learning, DL Architectures, Auto encoder, CNNs, RNNs, Transformers. Self Supervised Learning.
   - Total of 11-12 Lectures

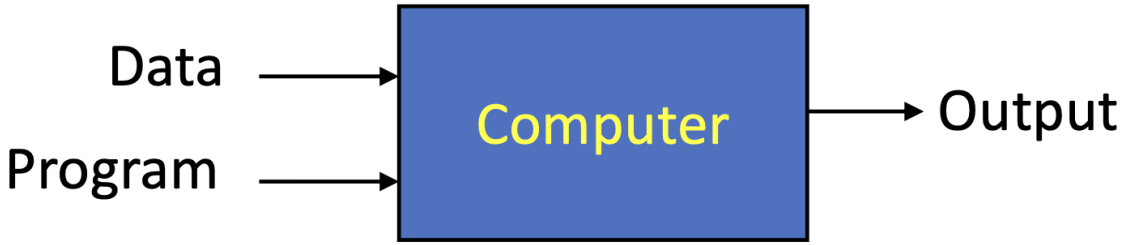
Course Evaluation

- Exams/Quiz (45%)
  - Quiz ($7.5 + 7.5$)
  - Mid (12)
  - Final (18)
- Class Work (20%)
  - Problem Solving (30-45 mins within class)
  - Best 80% out of what we do.
- HomeWorks/Assignments (20%)
  - Three Assignments
- Let us talk ML (10%)
  - Conceptual Dialogues with AI/Human (best 25 out of 30)
- Course Participation (5%)
  - Expect active participation in class, online etc.
- A max of change $\pm 5\%$, if required. Beyond will be informed.

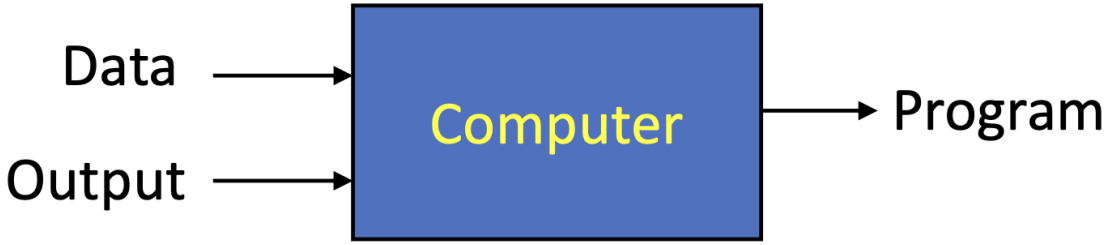
Role of ML in Software/Systems

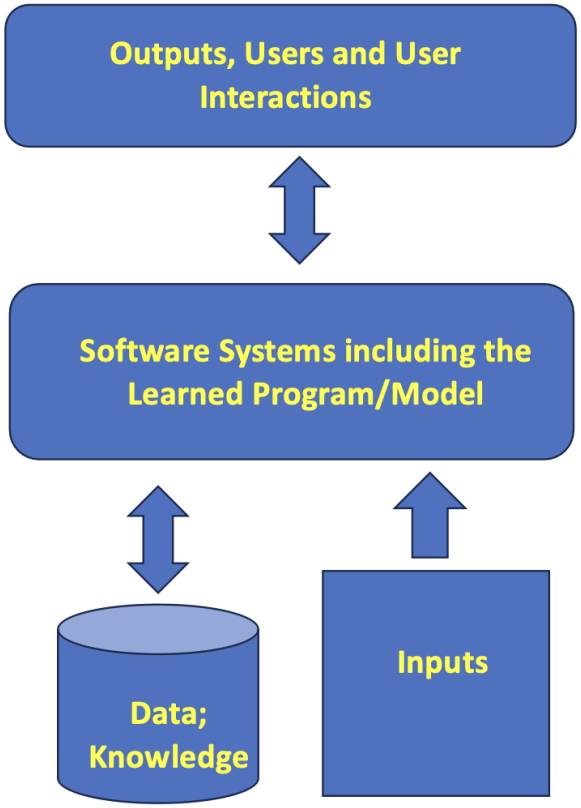
ML at the heart of todays AI systems

Traditional Programming



Machine Learning





Problem Space and Supervised Learning

- ML algorithms aims to learn/find $\mathbf{W}$ from the Data $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}$ $i = 1, \ldots, N$.
- Usually only
  - a portion of the data is used for "Training" (developing the solution or computing $\mathbf{W}$) and
  - the rest of the data is used for "Testing" (or measuring the performance.
- ML solution/program:

$$y = f(\mathbf{W}, \mathbf{x})$$

- $\mathbf{W}$ is the learnable parameters.

Learning Paradigms

- Supervised Learning
  - This is where we mostly focus on
- Unsupervised Learning
  - No supervision (eg. clustering)
- Self-Supervised Learning
  - Data provides supervisory signal
- Semi Supervised Learning
  - Partly supervised and partly unsupervised
- Reinforcement Learning
  - interact with environment and receive rewards

Representation is the key!

- Represent in the form of $\mathbf{x} \in R^d$
  - A good day to go for outing/games/picnic
  - Representing an email for spam classification
  - Problem: Orange vs Apple on a conveyor belt
- How?
  - Handcode or get from a sensor (thermometer)
  - Compute using statistics or signal processing
  - Compute using a Learned Solution (like a NN)

Type of Inputs and Outputs

- Inputs:
  - Vectors (also scalars etc.)
  - Sequences (of vectors)
- Output:
  - Regression (real number)
  - Classification (binary number or integer)
  - Structured Prediction (string, tree, graph etc.)

Nearest Neighbour Algorithm

- Nearest neighbor Algorithm for Classification
  - Classify as the label of the nearest neighbour.
- K-NN (K Nearest Neighbour) Algorithm
  - Classify based on the majority labels in the K neighbourhood.
- Questions:
  - What should be the value of K?
  - How do we know which sample is near?
  - What is the distance? Euclidean always?

Linear Classifier

- A simple Linear "Binary" Classifier
  - Classify as $sign(\mathbf{w}^T\mathbf{x})$
  - Either +ve or -ve. (Binary classification)
- Geometric Intuition
  - Which side of a line/plane/hyperplane
  - Partition input space as either + or -

Discussions

Q: $\mathbf{w}^T\mathbf{x}$ is a line passing through origin. How do we characterize "general" line?

- Ans1: $\mathbf{w}^T\mathbf{x} + b$
- Ans2: $\mathbf{w}'^T\mathbf{x}'$ where $\mathbf{x}'$ is an augmented vector. i.e., $\mathbf{x}' = [\mathbf{x}^T, 1]^T$. (There is notational convenience for this.)
- (Refer the usage of "bias" in neural networks. You will appreciate this better)
- In general, when we write as $\mathbf{w}^T\mathbf{x}$, often $\mathbf{x}$ is augmented and bias is absorbed.

Discussions

Q: We know, if we have two classes, we can classify as:

$$sign(\mathbf{w}^T\mathbf{x})$$

Assume we have three classes, how do we extend this decision making logic?
Write an algorithm to classify a sample into one of the four possible classes.

Problem-1

Consider a two class classification problem with positive examples as

$$\{[1,1]^T, [2,2]^T, [3,2]^T\}$$

and negative examples as

$$\{[1,-1]^T, [2,-2]^T, [3,-2]^T\}$$

We are interested in classifying $[4,1]^T$. What is the prediction for $K = 1, 3$?
What about for $[4.0]^T$

Problem-2

Consider a classification rule:
"Decide as A if $\mathbf{w}^T x \geq 0$; Else Class B"
Problem: for $\mathbf{w} = [1, 1, 1, 1]^T$, classify the following samples:

(i) $[1, 1, 1, 1]$ (ii) $[0, 1, 2, 3]^T$ (ii) $[-1, +1, -1, +1]$

Problem-3

What is the angle between the two lines characterized by

$$\mathbf{w}_1 = [1, 1]^T$$

$$\mathbf{w}_2 = [1, -1]^T$$

(a) 0 (b) 45 (c) 90 (d) 120

Summary

- Representation as a vector in R^d
- Learn a function $y = f(\mathbf{W}, \mathbf{x})$ from the data.
  - Notion of Training and Testing
- Two Simple Classification Schemes:
  - Nearest Neighbour Algorithm (K NN)
  - Linear Classification
  - $sign(\mathbf{w}^T\mathbf{x})$; Either +ve or -ve.

What Next:?

- We will use this week (until 10 Jan) for streamlining the Course:
  - New students
  - Teething problems in online mode.
- Topics:
  - Data Transformations
  - Performance Metrics of ML solutions.
- Recap of (do it yourself!):
  - Vectors and Matrices
  - Probability Distributions and Bayes Theorem
- Logistics:
  - Formal Details on Moodle and Shiksha
  - TAs to be on Board

References/Further Work

- Shoprt Videos:
  - How Can Data be Useful in Solving Problems? https://www.youtube.com/watch?v=8xniRSjRyCQ&feature=youtu.be https://tinyurl.com/ydsec8pa (10 mins)
  - https://www.youtube.com/watch?v=P92mkhzt6Hg&feature=youtu.be or https://tinyurl.com/4upj2fjs (6 mins)
- A good text book:
  - Mathematics of Machine Learning (pdf available) https://mml-book.github.io/
- More on Moodle and Gyan